

JEE MAIN REVISION MASTER: ELECTROSTATICS

2026 5 QUESTIONS 8.33% WEIGHTAGE ↓28.62%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 53 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Coulomb's Law, Electric Field & Dipole	39 PYQs	Forces & Fields
Very High	Gauss's Law & Electrostatic Potential	24 PYQs	Flux & Potential

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Coulomb's Law, Electric Field & Dipole	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$, $E_{\text{axial}} = \frac{2kp}{r^3}$, $E_{\text{equatorial}} = \frac{kp}{r^3}$, $\vec{\tau} = \vec{p} \times \vec{E}$, $U = -\vec{p} \cdot \vec{E}$
Gauss's Law & Electrostatic Potential	$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$, $E = \frac{\sigma}{2\epsilon_0}$ (sheet), $E = \frac{\lambda}{2\pi\epsilon_0 r}$ (wire), $\vec{E} = -\nabla V$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Coulomb's Law, Electric Field & Dipole (Very High Yield — 39 PYQs)

Core Principle: Electrostatic forces, electric field intensity due to point charges and dipoles, and dipole torque/energy in uniform fields.

■ **Caution / JEE Pitfall:** $E_{\text{axial}} = 2E_{\text{equatorial}}$ for a short dipole. $\vec{E}_{\text{equatorial}}$ points opposite to dipole moment $\vec{p}$.

■ **Speed Trick:** Work done in rotating dipole from θ_1 to θ_2 : $W = pE(\cos\theta_1 - \cos\theta_2)$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "An electric dipole of dipole moment $\vec{p}$ is placed in the uniform electric field $\vec{E}$. Then which of the following statements are..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Consider three point charges $-2Q, Q$ and $-Q$ and three surfaces $S_{(1)}, S_{(2)}$ and $S_{(3)}$ as shown in the figure. Match the entries of List..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Charges are uniformly spread on the surface of a conducting sphere. The electric field from the centre of sphere to a point outside the sphere..." [Ans: N/A]

3.2 Gauss's Law & Electrostatic Potential (Very High Yield — 24 PYQs)

Core Principle: Gauss's flux theorem, symmetry applications (infinite sheet, cylinder, sphere), and electric potential gradient $\text{vec}(\mathbf{E}) = -\text{vec}(\nabla)V$.

■ **Caution / JEE Pitfall:** Inside a charged hollow conducting sphere, electric field is ZERO, but potential is NON-ZERO and constant: $V_{\text{(inside)}} = (kq / R)$.

■ **Speed Trick:** Electric field lines are always perpendicular to equipotential surfaces and point in the direction of steepest potential decrease.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt..." **[Ans: N/A]**
- **[2026 Shift] Q#1 (MCQ):** "An electron falls through a distance 1.5 cm in 2.0×10^4 N/C from rest. The time taken to cover this distance in second is ____ ..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "Consider three point charges -2Q, Q and -Q and three surfaces $S_{(1)}$, $S_{(2)}$ and $S_{(3)}$ as shown in the figure. Match the entries of List..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt. The value of V at B is

- (A) 10 V
- (B) 20 V
- (C) 40 V
- (D) 60 V

Correct Answer: None

Solution Analysis: Work done in moving a charge from one point to another is related to potential difference by $W = q(V_B - V_A)$ Here, $W = 200\text{J}$ $q = 5\text{C}$ $V_A = -20\text{V}$ So, $200 = 5[V_B - (-20)]$ $200 = 5(V_B + 20)$ Divide both sides by 5 : $40 = V_B + 20$ $V_B = 20\text{V}$ So, the value of potential at point B is 20V Correct option: Option B.

Benchmark Problem 2: KCET 2026 (Q#2)

An electric dipole of dipole moment $\text{vec}(\mathbf{P})$ is placed in the uniform electric field $\text{vec}(\mathbf{E})$. Then which of the following statements are correct? Statement-I: The torque on the dipole is $\text{vec}(\mathbf{P}) \times \text{vec}(\mathbf{E})$ Statement-II: The potential energy of the dipole is $-\text{vec}(\mathbf{P}) \cdot \text{vec}(\mathbf{E})$ Statement-III: The net force on the dipole is non-zero.

- (A) I, II and III
- (B) I and II only
- (C) II and III only
- (D) I and III only

Correct Answer: None

Solution Analysis: Statement-I is correct. For an electric dipole placed in a uniform electric field, the torque acting on it is $\text{vec}(\tau) = \text{vec}(\mathbf{P}) \times \text{vec}(\mathbf{E})$ This torque tends to rotate the dipole so that $\text{vec}(\mathbf{P})$ becomes aligned with $\text{vec}(\mathbf{E})$. Statement-II is also correct. The potential energy of an electric dipole in a uniform electric field is $U = -\text{vec}(\mathbf{P}) \cdot \text{vec}(\mathbf{E})$ or $U = -PE\cos\theta$ where θ is the angle between $\text{vec}(\mathbf{P})$ and $\text{vec}(\mathbf{E})$. Statement-III is incorrect. In a uniform electric field, the positive and negative charges of the dipole experience equal and opposite forces. So the net force is zero.